

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: AUX DEFER in LTTPR Transparent
Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:

When operating in an LTTPR Transparent mode, the DP IN Adapter is responding to AUX transactions when it is Internal AUX Transaction. In those cases, the DP IN Adapter responds with DEFER if the timeout expires.

Benefits as a result of the changes:

Make the USB4 spec coherent

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
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None

An analysis of the hardware implications:
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None

An analysis of the software implications:
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None

An analysis of the compliance testing implications:
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None

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Actual Change

(a). Section 10.3.1 DisplayPort

Aux Transaction handling has two major aspects to attend when executed over the tunnel:

- AUX Transaction initiation operates in two ways:
 - DPTX Only – In this mode only DPTX initiates AUX Transactions. This mode is used when operating in LTTPR Non-Transparent mode. It is also used in LTTPR Transparent mode when not in the process of Link Training.
 - DPTX and DP OUT Adapter – In this mode both, the DPTX and the DP OUT Adapter, are initiating AUX Transactions. A DP OUT Adapter mainly initiate AUX Transactions as part of the autonomous and concurrent Link Training Process. This mode is used when operating in:
 - Non-LTTPR and, see Section 10.4.4.2.3
 - LTTPR Transparent Link Training, modes, see Section 10.4.4.2.3 and see Section 10.4.4.3.3 respectively.
 - 128b/132b LTTPR Link Training, see Section 10.4.10.3.
- Timeout and DEFER usage:
 - Allowed – A DP IN Adapter uses timeout to govern the AUX Link and uses AUX DEFER when timeout expires. This mode is used when operating in:
 - Non-LTTPR mode, see Section 10.4.4.2.1.
 - LTTPR Transparent, when addressing internal DPCD addresses, see Section 10.4.4.3.1
 - 128b/132b LTTPR Link Training, see Section 10.4.10.3.
 - Disallowed – A DP IN Adapter does not activate a timeout for AUX Responses, will not send DEFER and will not gate AUX Responses coming from DPRX. This mode is used when operating in LTTPR Non-Transparent mode and LTTPR Transparent when addressing noninternal DPCD addresses modes, see Section 10.4.4.3.1.

Table 10-2 summarizes the above modes of operations.

Table 10-2. DisplayPort Modes of Operation Over DisplayPort Tunneling

Mode	Link Training	AUX Transaction Initiation	AUX DEFER Usage
Non-LTTPR	Autonomous, Concurrent	DPTX and DP OUT Adapter	Allowed
LTTPR Transparent	Autonomous, Concurrent	DPTX and DP OUT Adapter	<u>Addressing internal DPCD: Allowed</u> <u>Addressing noninternal DPCD: Disallowed</u>

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LTTTPR Non-Transparent	DPTX Managed, Sequential	DPTX Only	Disallowed
128b/132b LTTTPR	DPTX Managed, Concurrent	Intra AUX Hop enabled: DPTX and DP OUT Adapter Intra AUX Hop disabled: DPTX only	Intra AUX Hop enabled: allowed Intra AUX Hop disabled: disallowed

(b). Section 10.4.4.3 LTTTPR Transparent Mode

10.4.4.3.1 AUX Timeout Timers

The AUX Response Timeout timer in a DP IN Adapter shall be set to 300 μ s when the AUX Transaction is an Internal AUX Transaction. For an External and Combined AUX Transactions, the AUX Response Timeout timer in a DP IN Adapter shall not be activated.

The AUX Reply Timeout timer in a DP OUT Adapter shall be set to 3.2 ms.

10.4.4.3.2 DP IN Adapter Requirements

A DP IN Adapter that receives an AUX Request shall classify the AUX Transaction as one of the following types:

- Internal AUX Transaction – AUX Request which targets only DPCD addresses that are defined as internal in Table 10-10.
- External AUX Transaction – AUX Request which targets only DPCD addresses that are not defined as internal in Table 10-10.
- Combined AUX Transaction – AUX Request which targets both Internal and External DPCD addresses. The AUX Response is initially generated by the DPRX and altered by the DP IN Adapter.

Note: A DP IN Adapter may assume that an AUX Write Request is not classified as a Combined Aux Transaction.

For External and Combined AUX Transactions, a DP IN Adapter shall send the AUX Request downstream to the DP OUT Adapter.

When a DP IN Adapter updates the DP OUT Adapter as the result of an External or Combined transaction, it shall perform the update only after receiving an AUX Response and before sending the AUX ACK response to the DPTX. The update shall be done by sending SET_CONFIG Packets to the DP OUT Adapter.

A DP IN Adapter shall not:

- Generate AUX DEFER, unless it is an Internal AUX Transaction.
- Gate any External or Combined AUX Request sent by the DPTX.
- Gate any AUX Response sent by the DPRX.